Business Intelligence Chapter 1 – Review Exercises

Exercise 1: Multiple Choice Questions (Fundamentals of BI)

Select the one best answer for each question.

1. **Which of the following best describes *Business Intelligence (BI)*?**
   a. A set of tools and processes for analyzing data to support better decision-making in organizations.
   b. A system used only for daily transaction processing in businesses.
   c. A methodology to program all managerial decisions automatically.
   d. An exclusive hardware technology for storing big data.

2. **Which of the following is *NOT* a typical benefit of implementing BI in a company?**
   a. Faster and more informed decision-making at various management levels.
   b. Identification of market trends and customer behaviors to gain competitive advantage.
   c. Significantly higher data entry workload and slower reporting processes.
   d. Improved strategic planning through a high-level overview of business performance.

3. **Which statement is true about the data sources used in BI solutions?**
   a. BI systems analyze only internal company databases and ignore external data.
   b. BI can utilize data from multiple sources, including internal databases and external, unstructured sources like social media.
   c. BI relies exclusively on real-time streaming data and does not use historical data.
   d. BI eliminates the need to consolidate data, since all data comes pre-integrated.

4. **In the context of data integration for BI, what does ETL stand for?**
   a. **Extract, Transform, Load** – the process of collecting data, converting it into a suitable format, and importing it into a data warehouse.
   b. Evaluate, Transfer, Log – the steps for evaluating data quality, moving data, and logging results.
   c. Enter, Transform, Link – the method of entering data, transforming it, and linking reports.
   d. Extract, Transfer, Learn – a cycle of extracting data, moving it elsewhere, and applying machine learning.

5. **Which of the following best defines a *Key Performance Indicator (KPI)*?**
   a. A quantifiable metric that reflects how well a key business objective is being achieved (e.g. profit margin or customer satisfaction rate).
   b. A type of database table that stores historical transaction records in a data warehouse.
   c. An automated software tool for extracting and cleaning data from sources.
   d. A visual dashboard used by executives, identical to a Business Intelligence report.

6. **What is the primary purpose of a *Data Warehouse* in BI?**
   a. To store integrated, historical data from various sources in a centralized repository for analysis and reporting.
   b. To run the company's day-to-day operational transactions (like an OLTP system).

c. To replace all databases in the organization with one single system.
d. To generate real-time sensor data and stream it directly to user dashboards.

Solution

1- a. BI is a set of tools and processes used to analyze data and support better decisions.

2- c. BI is meant to speed up reporting, not make it slower with more workload.

3- b. BI integrates internal and external sources like ERP systems and social media.

4- a. ETL stands for Extract, Transform, Load – it prepares and stores data for analysis.

5- a. A KPI is a measurable indicator that evaluates performance toward a goal.

6- a. A Data Warehouse stores integrated historical data for reporting and BI analysis.

Exercise 2

Provide brief answers (2–4 sentences each) to the following questions:

a. **Differentiate between *programmed (structured) decisions* and *non-programmed (unstructured) decisions*. Give an example of each in a business context.**

b. **What is the difference between a *Data Warehouse* and a *Data Mart*?** (Explain in terms of scope and purpose for each.)

c. **Briefly outline one or two key milestones in the historical evolution of Business Intelligence.** (For example, mention earlier decision-support systems or technologies that led to modern BI.)

Solution

Programmed vs Non-Programmed Decisions

- Programmed decisions are routine, repetitive, and based on predefined rules (e.g., automatic restocking when inventory is low).
- Non-programmed decisions are complex, unique, and require human judgment (e.g., choosing a marketing strategy for a new product).

b. Data Warehouse vs Data Mart

- A Data Warehouse is a central repository that collects all company data for global analysis.
- A Data Mart is a smaller version focused on one department, like sales or finance, for targeted analysis.

c. Historical Milestones of BI

- In the 1970s, DSS systems helped managers analyze data to support decisions.

- In the 1990s, BI evolved with data warehouses and OLAP tools, enabling more flexible and self-service analysis.

Exercise 3

Scenario: *RetailCo* is a regional retail company that operates a chain of supermarkets and an online store. The company has been collecting a large volume of data from its point-of-sale systems, customer loyalty programs, website transactions, and even social media feedback. Now, RetailCo's management wants to implement a **Business Intelligence solution** to improve its marketing strategies and inventory management decisions. They aim to better understand customer behavior, optimize stock levels, and increase sales using data-driven insights.

Tasks: Based on the scenario above, answer the following questions:

1. **Data Sources:** What data sources should RetailCo integrate into its BI system (data warehouse) to support analysis for marketing and inventory decisions? List **three** important data sources and briefly explain why each is relevant.
2. **Key Performance Indicators:** Suggest **two** Key Performance Indicators (KPIs) that RetailCo should track as part of this BI initiative, and explain what each KPI measures in this context. (Think of metrics that reflect marketing success or inventory efficiency.)
3. **BI Tools & Techniques:** Which BI tools or analysis techniques would you recommend to help RetailCo get useful insights from this data? Name **two** specific tools/techniques (for example: reporting dashboards, data visualization, OLAP, or predictive analytics) and briefly describe how each would assist RetailCo's managers in decision-making.

Solution

Interactive Dashboards & Data Visualization: RetailCo managers can use BI dashboards to monitor sales and stock levels in real time. A dashboard might show sales by store on a map, current inventory status, and alerts for low-stock items. Charts and graphs make it easier to spot trends and take quick action, especially in marketing and inventory control.

Predictive Analytics: By using tools like forecasting models or data mining, RetailCo can predict customer behavior and future product demand. For example, they can forecast which items will sell more next month or which customers may stop buying. These insights help the company plan ahead and act early, such as preparing stock or targeting offers.